Research Highlights

Entropic Collapse Model of Molecular Stability: A Self-Referential Thermodynamic Framework

1. **Novel Theoretical Framework**: We introduce a first-principles approach to molecular stability based on self-referential entropic collapse dynamics, providing a structural origin for entropy that traditional thermodynamics lacks.

2. **Collapse Entropy**: Our newly defined "collapse entropy" (E_collapse) quantifies structural degeneracy in information space, allowing for stability predictions directly from molecular topology without requiring empirical input.

3. **Quantum-Classical Bridge**: The framework naturally bridges quantum and classical regimes, unifying decoherence theory with macroscopic thermodynamics through the concept of information collapse.

4. **Predictive Power**: The Collapse Stability Index (CSI) predicts relative stabilities of isomers, conformers, and reaction pathways purely from structural information, offering a powerful tool for computational chemistry.

5. **Aromaticity Reinterpreted**: We provide a novel interpretation of aromaticity as a ψ-cycle resonance phenomenon, explaining its exceptional stability through minimized collapse entropy.

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